

Research V8R R3 Numerical Sensitivity Diagnostic for PID and Fuzzy PID Manipulator Control

Diagnostic supplement for frozen Research-v8R controllers

Scope and diagnostic questions

This report tests whether selected simulated responses change materially when the numerical time grid is refined. It uses four fixed controllers and five frozen cases. It is not an optimization study, not a controller-selection exercise, and not a confirmation study.

All 60/60 scheduled runs were attempted and completed. The analysis formed 40/40 finite complete pairs: 20/20 plant-half-step comparisons were stable, while coupled-half-step produced 6 sensitive and 14 stable comparisons, with 0 inconclusive comparisons and 0 baseline mismatches.

The primary conclusion is bounded. Holding the controller command sequence while halving only the plant integration step reproduced the original trajectories within preregistered practical tolerances for these cases. Halving the whole feedback-loop period changed six case-controller pairs enough to trigger at least one tolerance. This supports a local sampling-sensitivity diagnosis; it does not prove universal convergence or controller superiority.

The diagnostic questions are: (1) does plant-only refinement reproduce the archived response; (2) does coupled controller-and-plant refinement expose sample-rate sensitivity; and (3) how do arrival time, tracking error, positive mechanical work, squared torque and torque slew trade off in the frozen examples?

Model and evidence lineage

The simulated plant is the project two-link nonlinear manipulator with configuration-dependent inertia, Coriolis and centrifugal terms, gravity, viscous friction, payload, actuator limits and optional disturbances [1]. The manipulator formulation is consistent with the joint-space dynamics used in established robot-control studies [5]. Manual PID and Mamdani Fuzzy-PID provide the historical baselines [2,3], with fuzzy-set terminology following Zadeh's foundational definition [4]. Research V8 PID and Research V8 Fuzzy PID are the already frozen reliability controllers; R3 did not alter their parameters.

The evidence lineage is sealed generation -> independent statistical and trace packages -> this report. The builder authenticates the sealed generation's exact inventory and SHA-256 values, then recomputes report values from the raw diagnostic run, pairwise comparison, case-map and preregistration records. Generated narrative summaries are not numeric inputs.

The R3 source manifest hash begins fd0532d6152a407f. Its protocol is research-v8rs-r3, its generation evidence label is Research-v8R recovery evidence R3, FormalManifestEligible is true, and ConfirmationAdmitted is false. The immutable run and pairwise CSVs retain the authenticated nested row-field evidence label Research-v8R recovery evidence from the legacy runner. This is provenance nesting, not a different experiment. Independent media manifests must disclose both labels and bind the same source-manifest hash and the same four controller hashes.

Three profiles are compared. original-1ms uses a 1 ms controller period and 1 ms plant step. plant-half-step keeps the 1 ms controller period and held controller commands while using a 0.5 ms plant step. coupled-half-step uses a 0.5 ms controller period and 0.5 ms plant step, so it measures whole-loop sample-rate sensitivity rather than plant integration alone.

Cases profiles and tolerances

The five cases were chosen before outcome inspection by the frozen rules shown below. Three cases probe nominal, noisy/disturbed and long-travel reliability. Two matched cases replay the given-path and endpoint-only arrival commands for the same selected draw.

Case	Kind	Frozen selection rule
v8-validation-0015	reliability	lexically-first-planner-accepted-nominal-input
v8-validation-0009	reliability	lexically-first-planner-accepted-noise-disturbance-input
v8-validation-0093	reliability	largest-prescribed-joint-endpoint-travel-among-planner-accepted-long-travel-inputs-tie-lexical-scenario-id
v8-arrival-validation-draw-00247/path	path	largest-prescribed-endpoint-displacement-tie-lexical-pair-id
v8-arrival-validation-draw-00247/point	point	largest-prescribed-endpoint-displacement-tie-lexical-pair-id

A pair is practically stable only when every applicable preregistered metric remains within its threshold and categorical flags agree. Arrival is quantized on the native or common sampling grid; a missing value never counts as success.

Preregistered comparison	Tolerance	Unit
Maximum joint-state delta	0.002	rad
Maximum endpoint delta	0.002	m
Relative positive-work delta	0.01	fraction
Relative squared-torque delta	0.01	fraction
Relative mean-slew delta	0.05	fraction
Stable-arrival-time delta	0.01	s

The inherited reliability evaluator uses a final 0.5 s steady window, joint RMS limits of 0.02 rad, joint maximum limits of 0.05 rad and an end-effector maximum limit of 0.15 m. These criteria classify the frozen examples; R3 does not retune them.

Completeness and golden replay

Execution completeness is exact: 60 scheduled, 60 attempted and 60 completed runs. The pairwise table contains 40 comparisons, all 40 of which are finite and complete. There are 0 baseline mismatches.

Each original-1ms run is checked against its frozen golden record before it can anchor a refinement pair. This test protects evidence identity: a numerically complete refinement cannot be interpreted if its reference replay no longer matches the archived controller-case contract. Golden replay is a lineage check, not evidence that the underlying physical model is exact.

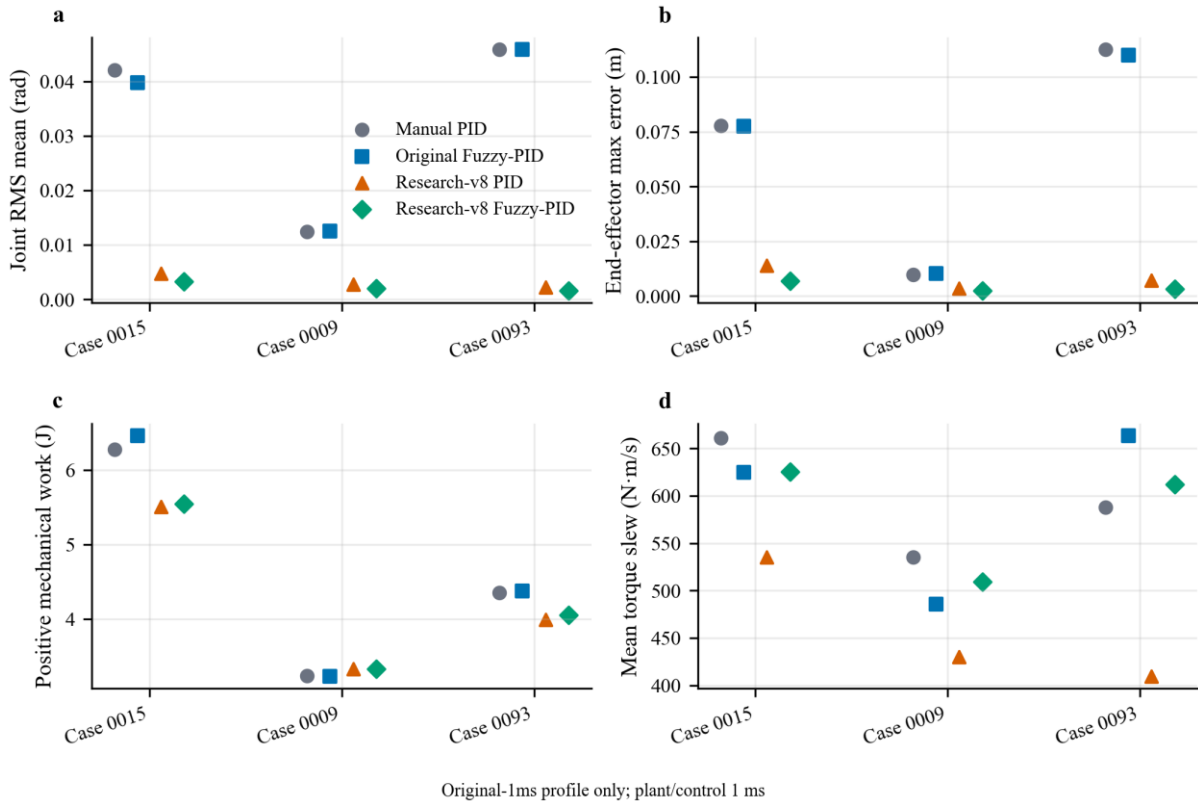


Figure 1. Original-grid tracking and reliability metrics for the three frozen reliability cases. Every case-controller row remains visible; the plot is descriptive, not a ranking over a population.

The nominal and noisy/disturbed cases are not interchangeable with the long-travel case. A controller can satisfy the conjunction in one case and fail it in another. Accordingly, the figure retains case identity rather than averaging five heterogeneous tasks into a single success rate.

Reliability traces

The state-error traces show the original grid beside both refinements for the representative long-travel reliability case v8-validation-0093. The three selected reliability cases remain separately tabulated in Figure 1. plant-half-step curves visually coincide because the measured differences are orders of magnitude below the practical limits. coupled-half-step curves also look close, although the relative torque-slew metric crosses its preregistered threshold in this case.

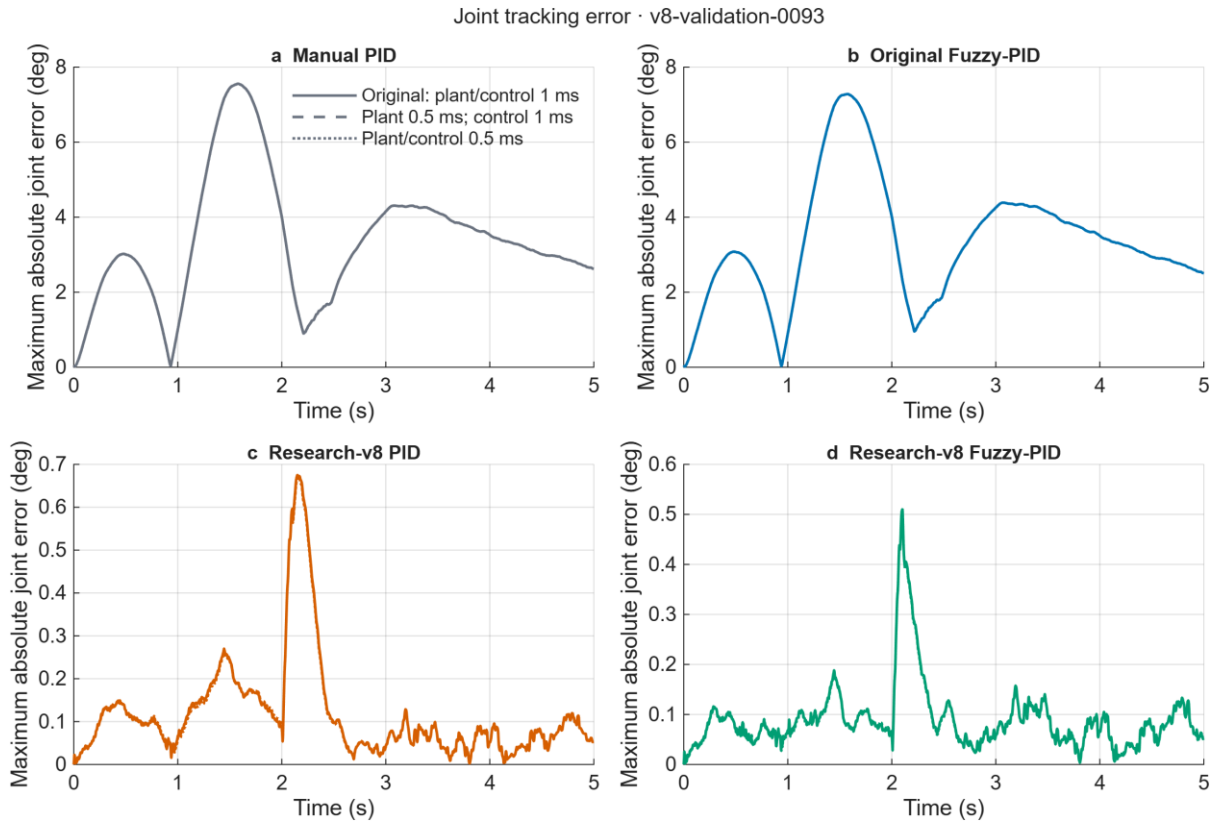


Figure 2. Recorded maximum absolute joint-error histories for long-travel reliability case v8-validation-0093 and every frozen controller across three profiles. Each panel uses its own y-axis scale to reveal within-controller profile differences. Figure 1 reports joint RMS, not the maximum error plotted here; the two figures answer different questions.

On the original grid, the research controllers satisfy ReliableSuccess in all three selected reliability cases. Manual PID fails the nominal and long-travel cases; Original Fuzzy PID fails the long-travel case. These are five-case diagnostic observations, not estimates of general controller success probability.

Given path arrival

The given-path task retains its prescribed Cartesian timing. Stable arrival time is the recorded output of the frozen arrival evaluator; R3 compares the recorded metric without changing the command generator or controller. All four original-grid path runs record SafeArrival=1.

Controller	Arrival s	Positive work J	Squared torque (N m) ² s	Mean slew N m/s
Manual PID	5.801	2.9460	660.51	15.29
Original Fuzzy PID	4.712	3.0687	670.17	15.71
Research V8 PID	2.659	3.1158	726.86	3.04
Research V8 Fuzzy PID	2.659	3.1158	726.86	3.04

Research V8 PID and Research V8 Fuzzy PID reach the recorded stable-arrival condition at 2.659 s, earlier than Manual PID at 5.801 s and Original Fuzzy PID at 4.712 s in this selected path case. Their positive mechanical work is about 3.116 J, slightly above the two historical controllers in this case. This is a paired case description, not a universal speed claim.

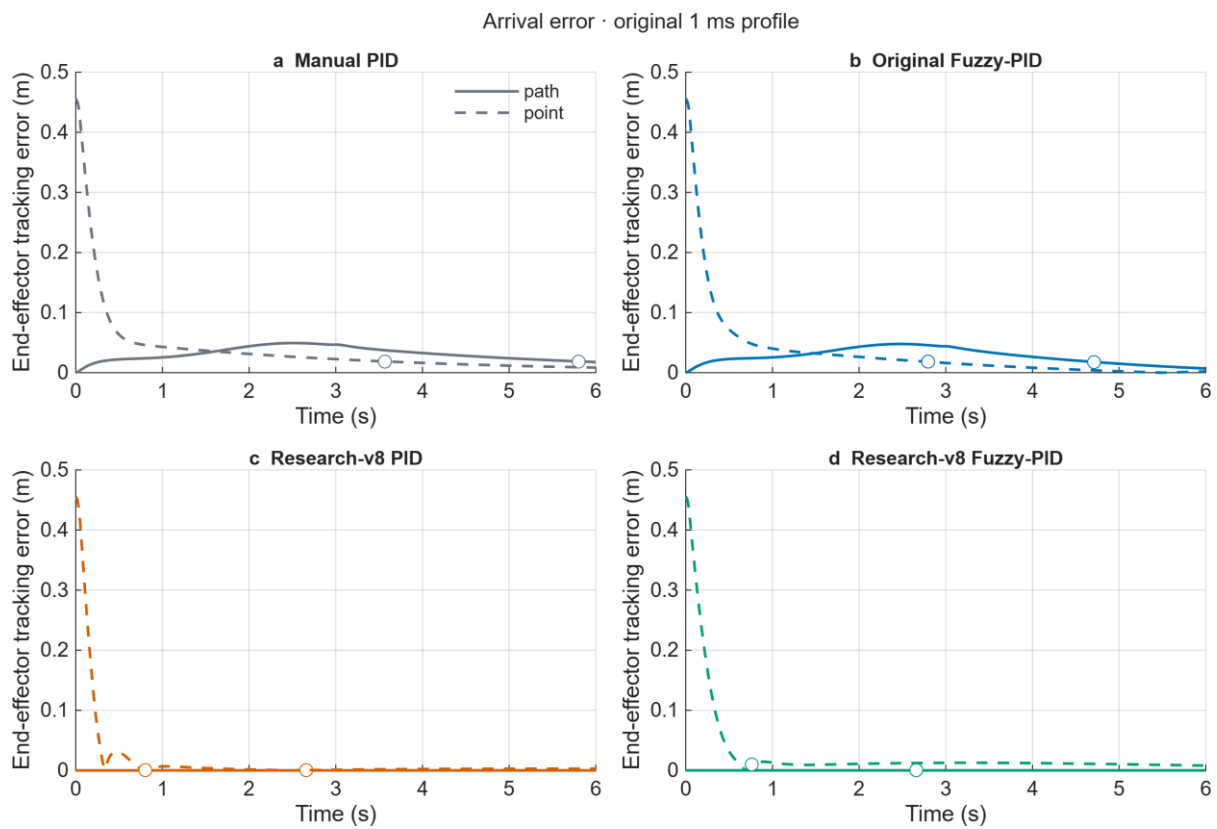


Figure 3. Recorded end-effector arrival-error histories on the original-1 ms profile only for the selected given-path and endpoint-only commands. Open circles mark the recorded stable-arrival times.

Endpoint only arrival

The endpoint-only task supplies the target without requiring the intermediate Cartesian path. All four original-grid point runs record SafeArrival=1. The task definition therefore permits a faster transient but does not make that transient directly equivalent to timed path tracking.

Controller	Arrival s	Positive work J	Squared torque (N m) ² s	Mean slew N m/s
Manual PID	3.565	4.5023	770.83	94.61
Original Fuzzy PID	2.793	4.9824	786.80	111.92
Research V8 PID	0.801	5.7612	847.31	79.41
Research V8 Fuzzy PID	0.760	3.0939	833.60	99.65

In this selected point case, Research V8 Fuzzy PID has the shortest recorded stable-arrival time, 0.760 s, and the lowest positive mechanical work, 3.0939 J. Research V8 PID records 0.801 s and 5.7612 J. Manual PID and Original Fuzzy PID take 3.565 s and 2.793 s, respectively. These outcomes expose a controller-specific time-work tradeoff within one matched endpoint pair.

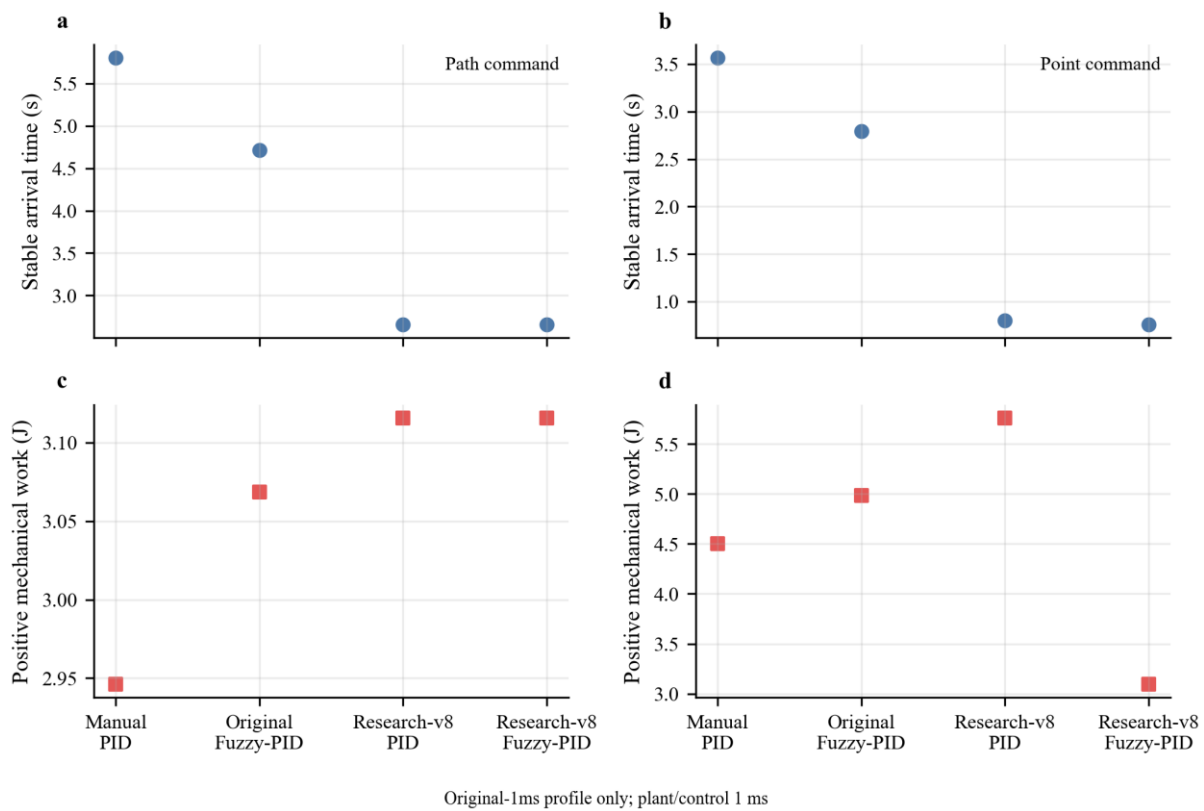


Figure 4. Original-grid stable-arrival time and positive mechanical work for the matched path and point cases. Lower values are not collapsed into a single score.

Actuation costs

Positive mechanical work is the time integral of positive applied joint power and is reported in J. It is not electrical energy because motor efficiency, drive loss and regeneration are not modelled. Squared torque integral is reported in $(\text{N m})^2 \text{ s}$. Mean torque slew is the mean across joints of RMS applied-torque derivative and is reported in N m/s .

The three metrics answer different questions. Positive work captures one-sided mechanical delivery; squared torque emphasizes sustained magnitude; slew emphasizes rapid command changes. A controller can improve one while worsening another. The point case illustrates this: Research V8 Fuzzy PID records less positive work than Research V8 PID, while their squared-torque and slew ordering does not produce an all-metric dominance statement.

Cumulative positive work · v8-validation-0093

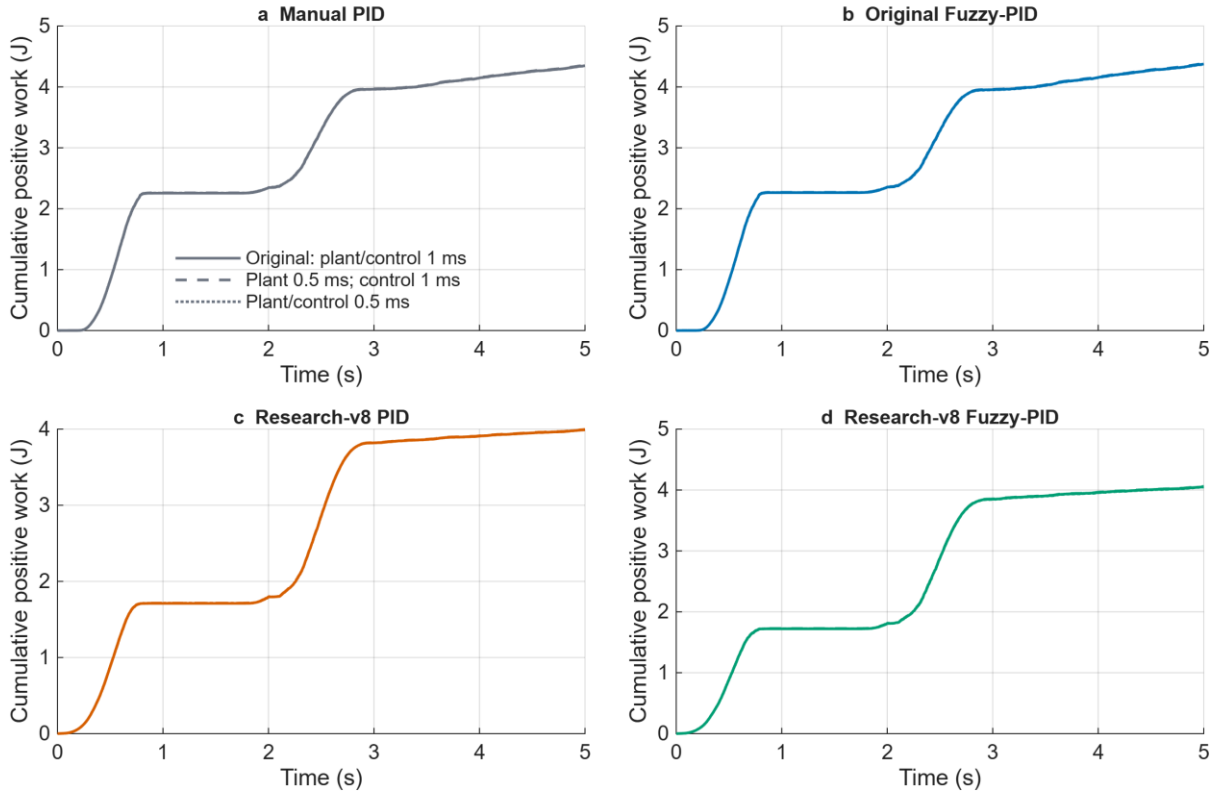


Figure 5. Recorded cumulative positive mechanical work for the long-travel reliability case v8-validation-0093 under all three profiles. Applied-torque traces are not displayed.

The work differences between original-1ms and plant-half-step are numerically negligible in all selected pairs. coupled-half-step remains close in absolute trajectory shape, but six relative torque-slew differences exceed the 5% preregistered limit. Categorical flags agree in all 40 refinement pairs. That pattern is consistent with feedback sampling dependence; it does not by itself reveal which controller component is causal.

Sensitivity results

plant-half-step: 20/20 comparisons were stable. coupled-half-step: 6 sensitive and 14 stable. Across both refinements there were 0 inconclusive comparisons and 0 baseline mismatches.

Sensitive coupled-half-step case	Controller
v8-validation-0009	Original Fuzzy PID
v8-validation-0009	Research V8 Fuzzy PID
v8-validation-0093	Manual PID
v8-validation-0093	Original Fuzzy PID
v8-validation-0093	Research V8 PID
v8-validation-0093	Research V8 Fuzzy PID

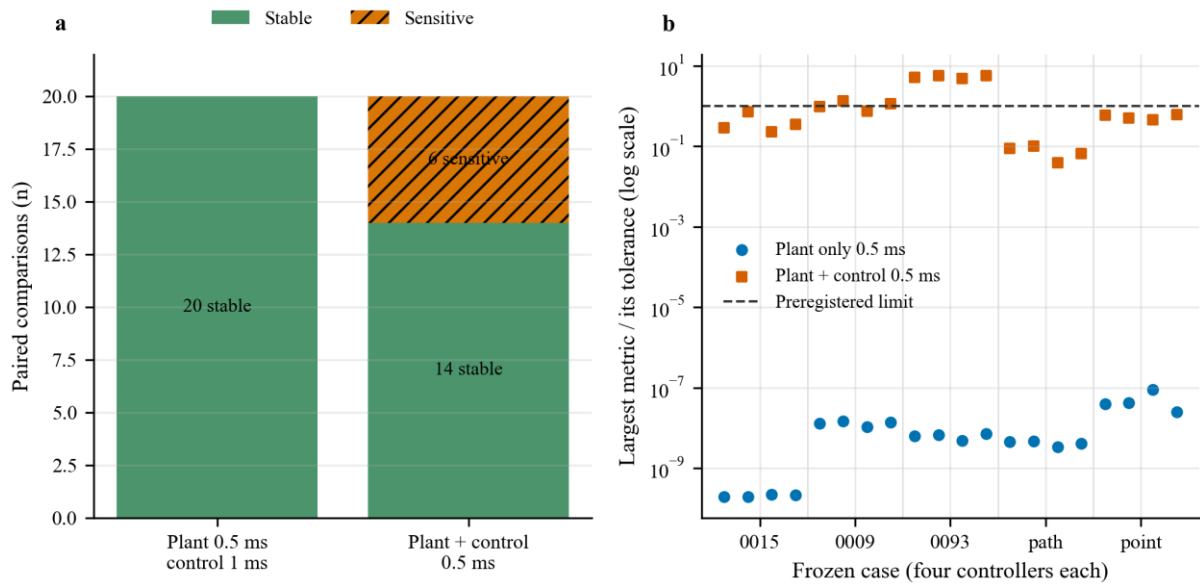


Figure 6. Pairwise practical-stability disposition by profile. Counts use all 40 preregistered refinement pairs.

The two Fuzzy-PID variants are sensitive in the noisy/disturbed case v8-validation-0009. All four controllers are sensitive in the long-travel case v8-validation-0093. These flags localize sampling sensitivity; they do not identify a unique physical cause. The arrival cases and nominal reliability case remain stable under both refinements.

Limitations and mechanism hypotheses

The study is intentionally small and diagnostic: five frozen inputs, four fixed controllers and two refinements against one original grid. It cannot estimate population reliability, validate hardware behaviour, or identify a globally sufficient simulation step. Practical thresholds are engineering decision limits, not mathematical error bounds [6,7].

The stable plant-half-step results support the narrow hypothesis that the archived 1 ms plant integration is not the dominant numerical source for these selected responses when commands are held at 1 ms. The coupled-half-step flags support a second hypothesis: controller update period and feedback sampling can affect the classified response in noisy/disturbed and long-travel cases. Both are mechanism hypotheses bounded to the simulated cases.

No component-level attribution is made. Derivative estimation, fuzzy gain scheduling, disturbance sampling, anti-windup state and discrete event timing could each contribute, but R3 does not isolate them. A causal follow-up would vary one discrete subsystem at a time while holding random draws, reference inputs and controller parameters fixed.

The report also avoids a common interpretation error: earlier arrival does not imply lower actuation cost, and a lower mechanical-work value does not imply lower electrical energy. The retained path and point tasks should remain separate in any future controller optimization.

Conclusion reproducibility and references

R3 completed its full frozen diagnostic without missing runs: 60/60 executions and 40/40 finite complete refinement pairs. Plant-only half-step refinement was stable in all 20 pairs. Coupled half-step refinement was sensitive in 6 pairs and stable in 14. The result supports the archived simulation for the tested plant-only refinement and identifies bounded whole-loop sampling sensitivity that should be addressed before stronger reliability claims.

The controllers were fixed throughout. No parameters were optimized, no controller was admitted, and no confirmation dataset was opened or rerun. Arrival and actuation comparisons describe the selected path and point examples only.

Reproducibility. Protocol research-v8rs-r3; evidence label Research-v8R recovery evidence R3; source-manifest SHA-256 fd0532d6152a407feb78f367d82ee1ad370ac59df1a94a16e108b183c352dd79; implementation commit c0d6a4b9f72a92d8a529878166751575ece54802. The sealed generation, independent figure manifests, independent video manifests, report source and build manifest bind every delivered artifact by SHA-256.

Video evidence

[Given-path diagnostic replay](#)

[Endpoint-only diagnostic replay](#)

References

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